Unit: Unit 8: Quadratics **Name:** ____________

Topic: Converting Forms (Completing the Square) **Date:** ________

Foundation (Jalapeno)

Q1. Convert $x^2 + 6x + 8$ to vertex form

(A) $(x+3)^2 - 1$
(B) $(x+3)^2 + 1$
(C) $(x+4)^2 - 8$
(D) $(x+4)^2 + 8$
(E) $(x+2)^2 - 4$

Q2. Complete the square for $x^2 + 4x$

(A) $x^2 + 4x + 4$
(B) $x^2 + 4x - 4$
(C) $(x+2)^2$
(D) $(x+2)^2 - 4$
(E) $(x-2)^2$

Q3. What is the vertex of the parabola $x^2 + 2x - 6$ in vertex form?

(A) $(-1, -7)$
(B) $(-1, 7)$
(C) $(-2, -6)$
(D) $(-2, 6)$
(E) $(-3, -9)$

Q4. A stock's price is modeled by $p(x) = x^2 + 5x - 3$. What is the vertex form of this equation?

(A) $p(x) = (x+2.5)^2 - 16.25$
(B) $p(x) = (x+2.5)^2 + 16.25$
(C) $p(x) = (x+3)^2 - 12$
(D) $p(x) = (x-2.5)^2 + 16.25$
(E) $p(x) = (x+1)^2 - 4$

Q5. A company's profit is given by $P(x) = x^2 + 10x - 5$. Convert this to vertex form.

(A) $P(x) = (x+5)^2 - 30$
(B) $P(x) = (x+5)^2 + 30$
(C) $P(x) = (x+4)^2 - 21$
(D) $P(x) = (x-5)^2 - 30$
(E) $P(x) = (x+2)^2 - 9$

Q6. Complete the square for $x^2 - 8x$

(A) $(x-4)^2 - 16$
(B) $(x-4)^2 + 16$
(C) $(x+4)^2 - 16$
(D) $x^2 - 8x + 16$
(E) $(x+2)^2 - 4$

Q7. The equation $x^2 + 12x + 20$ can be rewritten as

(A) $(x+6)^2 - 16$
(B) $(x-6)^2 + 16$
(C) $(x+10)^2 - 80$
(D) $(x-10)^2 + 80$
(E) $(x+2)^2 - 4$

Q8. What is the value of c in $x^2 + 4x + c$ if the vertex is $(-2, -4)$?

(A) -4
(B) -2
(C) 0
(D) 4
(E) 8

Open Ended Questions (FRQ)

Q9. A business's revenue is modeled by $R(x) = x^2 + 6x$. Complete the square to find the vertex form of this equation. Show all steps.

Q10. A tax consultant uses the equation $T(x) = x^2 + 10x + 25$ to model a client's tax burden. Convert this equation to vertex form and identify the vertex. Show all steps.

Chef's Tips

- Make sure students understand that completing the square involves creating a perfect square trinomial
- Remind students to identify the vertex when given a quadratic equation in vertex form
- Encourage students to apply these concepts to real-world problems in business and finance